

MD ZAKI HUSSAIN

AWS Data Engineer | Python, SQL, PySpark, AWS, Terraform

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SUMMARY

AWS Data Engineer within the selective TCS Prime track, specializing in zero-to-one data platform architecture, distributed computing (PySpark/Glue), and modern lakehouse formats (Apache Hudi) for Aegon UK. Expert at engineering scalable batch/near-real-time pipelines and dimensional warehouses from scratch, with a proven track record of executing complex platform migrations, runtime upgrades, and high-impact production release cycles.

EXPERIENCE

Tata Consultancy Services (TCS Prime) – AWS Data Engineer @ AEGON UK – Kolkata, India

Apr 2025 – Present

- **Zero-to-One Warehouse Architecture:** Spearheaded the end-to-end design of an enterprise data warehouse, building a generic, JSON-config-driven AWS Glue framework that automated ingestion and systematically replaced high-latency legacy SQL Server stored procedures.
 - **Lakehouse Engineering & CDC Modelling:** Built a robust data reconciliation engine utilizing **Apache Hudi** and PostgreSQL; implemented runtime hashing and optimized Spark joins/anti-joins to process SCD Type 1/2 dimensions, orchestrating full execution via AWS Step Functions and EventBridge rules.
 - **High-Scale Performance Tuning:** Optimized a critical 500M+ record PostgreSQL pipeline, cutting execution runtime from **20+ minutes to 4–5 minutes** by eliminating auto-bound overhead and deploying targeted SQL pushdown, interval filtering, and tuned JDBC partitioning.
 - **Production Operations & Modernization:** Directed 10+ production releases via Jenkins, serving as the primary resource for debugging live bottlenecks (Postgres locks, data duplication, DMS lags); upgraded Lambda microservices to Python 3.12 and leveraged **Amazon Q / Kiro AI** to accelerate codebase validation.
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SKILLS

Cloud & Lakehouse Architecture: AWS (Glue, Lambda, Step Functions, EventBridge, DMS, Athena, DynamoDB, RDS, IAM, CloudFormation), Apache Hudi, CDC, SCD Type 0, 1 & 2, Dimensional Modelling

Data Engineering & Languages: Python, C++, SQL (PostgreSQL, MS SQL Server), PySpark, DSA

DevOps, CI/CD & AI Tools: Terraform, Jenkins, CloudWatch, Amazon Q, Kiro IDE, VS Code

EDUCATION

Nalanda College of Engineering | Chandi, Bihar | *B.Tech in Computer Science & Engineering* | 2020 – 2024

- **CGPA:** 8.13 / 10 | **Core Focus:** Highest marks in DBMS, OS, and CN. Core DSA.

CERTIFICATIONS

Elite / Core Certification in Professional Communication & Soft Skills | NPTEL | [2022]